

ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

USEFUL PROPERTIES OF THE GAUSSIAN

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Closure Under Linear Transformation

Lemma. If $X \sim N(\mu, \Sigma)$ in $\mathbb{R}^d$ and A is a $k \times d$ matrix, then AX is a Gaussian in $\mathbb{R}^k$.

What are the mean and covariance of AX ?

Back to Sensor Example

Sensor readings $X_1, \dots, X_d$: model by a Gaussian $N(\mu, \Sigma)$.

- 1 How to set μ and Σ ?
- 2 What is the distribution of sensor X_i ?
- 3 What is the joint distribution of a set of sensors $S \subset [d]$?

Distribution Under Conditioning

- 4 Suppose sensors S fail. What is the conditional distribution of X_S given the readings X_T of the remaining sensors $T = [d] \setminus S$?

Lemma. The conditional distribution $X_S | X_T = x_T$ is Gaussian, with mean $\mu_S + \Sigma_{ST}\Sigma_{TT}^{-1}(x_T - \mu_T)$ and covariance $\Sigma_{SS} - \Sigma_{ST}\Sigma_{TT}^{-1}\Sigma_{TS}$.